

Remote User Authentication and Device Recognition System

A Multi-Layered Security Approach for Preventing Unauthorized Access

Introduction

This project presents a modern security system designed to protect user accounts from unauthorized access, especially from remote or unknown devices. Traditional authentication systems rely heavily on passwords alone, which are often vulnerable to phishing, brute force attacks, and data breaches. This system improves on that model by introducing an additional layer of device recognition, ensuring that access is granted not only based on identity but also on trusted hardware.

System Architecture

The system is built using a client-server architecture. The frontend (client) is responsible for collecting user input, generating a unique device identifier, and communicating with the backend. The backend (server) processes authentication logic, manages device authorization, and enforces access control policies. This separation ensures scalability, maintainability, and better security enforcement.

Core Security Components

1. Authentication Mechanism

The system uses bcrypt hashing to secure user passwords. When a user registers, their password is transformed into a hashed value that cannot be reversed. During login, the entered password is compared against the stored hash. This ensures that even if the database is compromised, user passwords remain protected.

2. Device Recognition

A unique Device ID (UUID) is generated on the client side and stored in the browser's localStorage. This ID acts as a digital fingerprint for the device. Only devices that have been previously registered and stored in the system's trusted list are allowed access.

3. Monitoring and Threat Detection

The system actively monitors login attempts. If a valid login is attempted from an unrecognized device, the system blocks access and logs the event. This feature helps identify suspicious activities and prevents unauthorized users from gaining entry even with correct credentials.

Workflow Explanation

Registration Phase

During registration, the user provides their email and password. A unique device ID is generated and sent to the server along with the credentials. The server hashes the password and stores both the hash and the device ID in a trusted device list.

Recognized Login Phase

When the user logs in from a recognized device, the system verifies both the password and the device ID. If both checks pass, access is granted and a session is created.

Unauthorized Access Attempt

If a login attempt is made from a new or unrecognized device, the system detects the mismatch in device ID. Even if the password is correct, access is denied, and the attempt is flagged for security monitoring.

Technology Stack

The system is built using a JavaScript-based stack. The backend uses Node.js and Express.js for handling server-side logic and API routing. Bcrypt is used for password security, while CORS ensures safe communication between frontend and backend.

The frontend is developed using HTML5, CSS3, and JavaScript. It handles user interaction, form submission, and device ID generation. Data exchange between the client and server is done using JSON.

Limitations and Future Improvements

Although effective, the system has some limitations. Currently, it uses in-memory storage, which means data is lost when the server restarts. In a real-world implementation, this should be replaced with a database such as MongoDB or PostgreSQL.

Future improvements include integrating Two-Factor Authentication (2FA), implementing email or SMS-based OTP verification for new devices, and adding geo-location tracking to detect unusual login locations.

Conclusion

This system demonstrates how combining multiple layers of security can significantly improve protection against unauthorized access. By integrating password authentication with device recognition and monitoring, the system provides a more robust and intelligent security solution suitable for modern web applications.